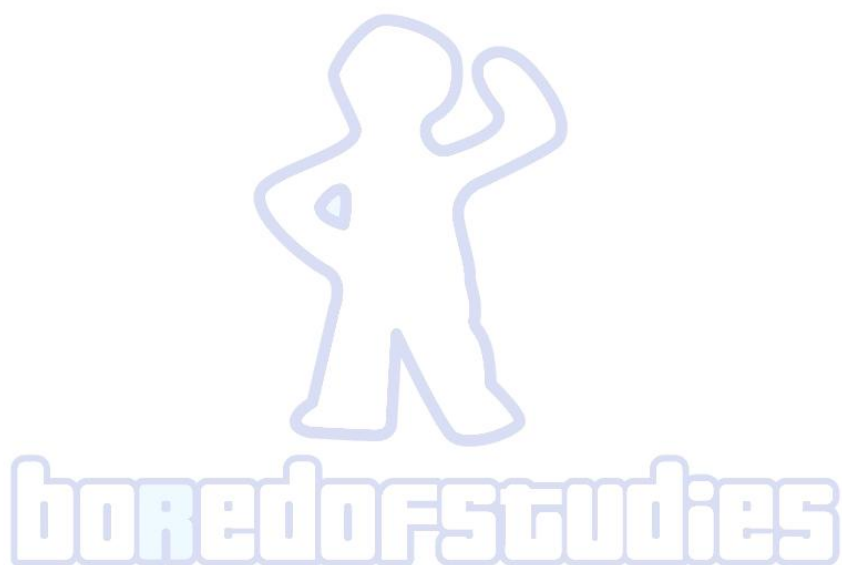




2018 Bored of Studies Trial Examinations

Mathematics Extension 2

8th October 2018

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A reference sheet has been provided.
- Show all necessary working in Questions 11 – 16.

Total Marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 6 – 17

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What is the limiting value of the following product?

$$\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right) \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right) \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8}\right) \left(\cos \frac{\pi}{16} + i \sin \frac{\pi}{16}\right) \dots$$

- (A) -1
(B) $-i$
(C) 1
(D) i
- 2 Which of the following parametric coordinates could describe the rectangular hyperbola $xy = 1$, for all possible values in its domain?
- (A) $(\cos \theta, \sec \theta)$
(B) $(\cos \theta, \cot \theta)$
(C) $(\tan \theta, \sec \theta)$
(D) $(\tan \theta, \cot \theta)$
- 3 A real-coefficient polynomial $P(x)$ has all roots with even multiplicity. Which of the following statements is always true?
- (A) $P(x)$ must be non-negative for all real x
(B) $P(x)$ must have both positive and negative values
(C) $P'(x)$ must be non-negative for all real x
(D) $P'(x)$ must have both positive and negative values

- 4 Let $f(x)$ be a continuous and differentiable function over all real x . The graph of $y = f(x)$ has n_1 stationary points and n_2 x -intercepts where $n_1 \geq n_2$.

The graph of $y = [f(x)]^2$ has n stationary points. Which of the following statements is always true?

- (A) $n = n_2$
- (B) $n \geq n_2$
- (C) $n = n_1 + n_2$
- (D) $n \geq n_1 + n_2$

- 5 Consider the points A , B and C on the first quadrant of the Argand diagram, which represent the complex numbers z_1 , z_2 and z_3 respectively. Suppose that $z_2 = z_1 + z_3$ and

$$\arg\left(\frac{z_1 - z_3}{z_1 + z_3}\right) = \arg\left(\frac{z_1}{z_3}\right) = \frac{\pi}{2}.$$

What is the most accurate description of the quadrilateral $OABC$?

- (A) Parallelogram
- (B) Rhombus
- (C) Rectangle
- (D) Square

- 6 Let P , Q and R be distinct points on the Argand diagram with equal modulus, represented by complex numbers p , q and r respectively.

If $r = 1$, which of the following conditions ensures that $\triangle PQR$ is equilateral?

- (A) $p + q = -1$
- (B) $p + q = 0$
- (C) $p + q = 1$
- (D) $p + q = 2$

- 7 Let $P(x)$ and $Q(x)$ be real-coefficient polynomials without any common factors. Suppose that

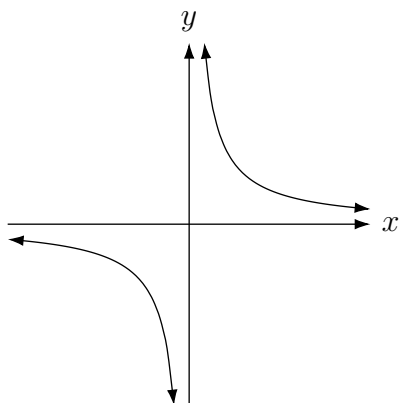
$$\int \frac{P(x)}{Q(x)} dx = x^2 + \ln |(x - a_1)(x - a_2)(x - a_3) \dots (x - a_n)| + C,$$

where $a_1, a_2, a_3, \dots, a_n$ are constants

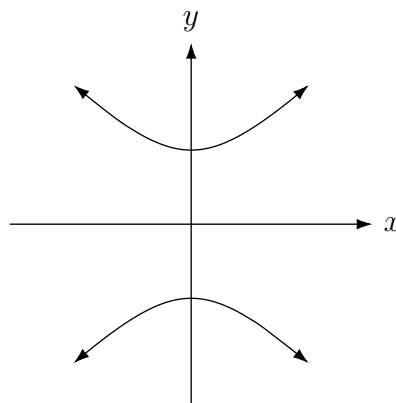
What is the degree of $P(x)$?

- (A) $n - 1$
 (B) n
 (C) $n + 1$
 (D) $n + 2$
- 8 Which of the following represents the graph of the locus of $||z - \sqrt{i}| - |z + \sqrt{i}|| = k$, for some $k > 0$?

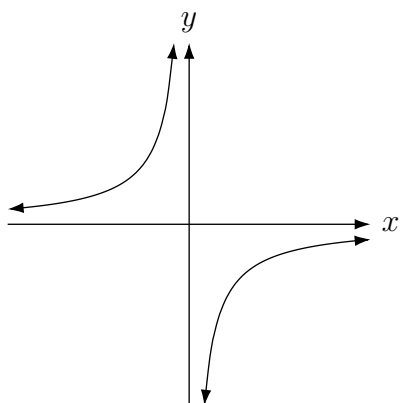
(A)



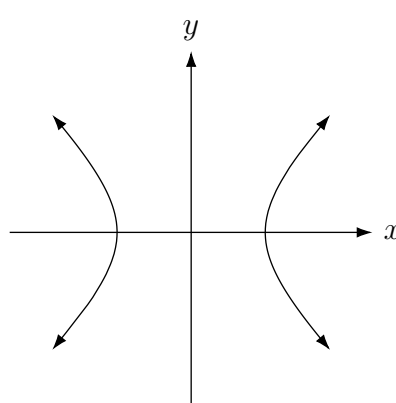
(C)



(B)



(D)



- 9** Let ω be a cube root of unity. Which of the following is the value of

$$(1 + \omega)(1 + \omega^2)(1 + \omega^3) \dots (1 + \omega^{2018})?$$

- (A) 2^{672}
(B) 2^{2016}
(C) $2^{671}(1 + i\sqrt{3})$
(D) $2^{671}(1 - i\sqrt{3})$
- 10** Consider n evenly-spaced points on a circle. Every point is connected to all other points, forming a number of intersecting chords. How many times do the chords intersect each other at most?

- (A) $\binom{n}{1}$
(B) $\binom{n}{2}$
(C) $\binom{n}{3}$
(D) $\binom{n}{4}$

Section II

90 marks

Attempt Questions 11—16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available. In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

(a) Evaluate $\int_0^1 \frac{dx}{2 + \sqrt{1+x} + \sqrt{1-x}}$. **3**

(b) Consider the curve $\mathcal{C}$ defined parametrically by

$$x = \frac{3t}{1+t^3} \quad \text{and} \quad y = \frac{3t^2}{1+t^3}.$$

(i) Show that the Cartesian equation of $\mathcal{C}$ is $x^3 + y^3 = 3xy$. **1**

(ii) What is the limiting value of t so that $x \rightarrow \infty$? **1**

(iii) By considering $x + y$, deduce that $y = -x - 1$ is an asymptote. **1**

(iv) Find the critical points of $\mathcal{C}$. **3**

(v) Hence, sketch the graph of $\mathcal{C}$ showing the critical points and asymptote. **2**

(c) Suppose $f(x)$ is a monotonic function for $a \leq x \leq b$, where $f(a) = c$ and $f(b) = d$.

(i) Show that **2**

$$\int_c^d f^{-1}(x) dx = \int_a^b x f'(x) dx.$$

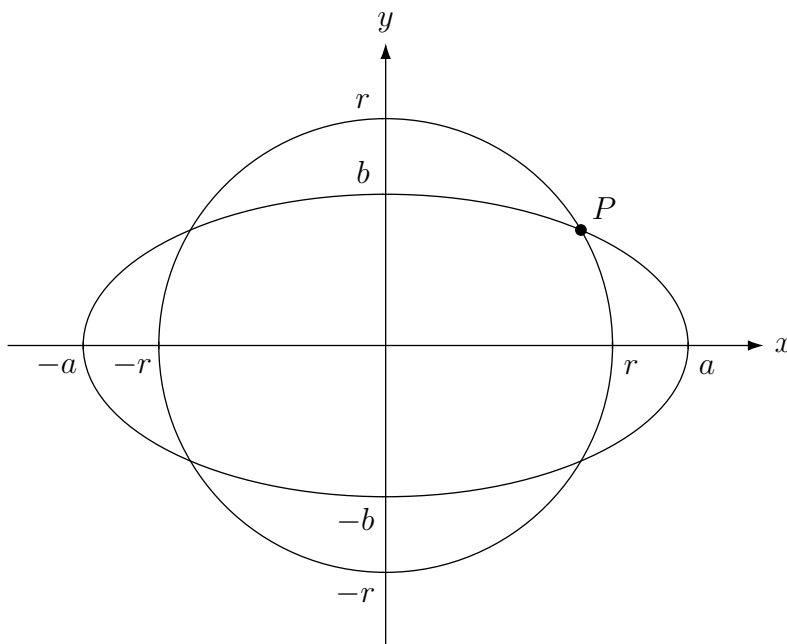
(ii) Hence, or otherwise, show that **2**

$$\int_a^b f(x) dx + \int_c^d f^{-1}(x) dx = bd - ac.$$

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) Consider the ellipse and circle with Cartesian equations $x^2 + y^2 = r^2$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b < r < a$ so that the circle intersects the ellipse at 4 points.



- (i) Show that the circle intersects the ellipse in the first quadrant at the point

$$P \left(a\sqrt{\frac{r^2 - b^2}{a^2 - b^2}}, b\sqrt{\frac{a^2 - r^2}{a^2 - b^2}} \right).$$

- (ii) Let θ be the acute angle between the circle and the ellipse at P . Show that

$$\tan \theta = \frac{\sqrt{(a^2 - r^2)(r^2 - b^2)}}{ab}.$$

- (iii) Suppose the axes of the ellipse are fixed, and the radius of the circle can vary. Show that the maximum value of θ occurs when $r = \sqrt{\frac{a^2 + b^2}{2}}$.

Question 12 continues on page 8

Question 12 (continued)

- (b) Define the polynomial $P(x) = x^3 - x^2 + k$, where k is real, such that $P(x)$ has exactly two non-real roots $a \pm ib$ and one real root γ .

- (i) Show that

$$(a^2 + b^2) [k - (a^2 + b^2)^2] = -k^2.$$

3

- (ii) Explain why $a^2 + b^2$ is the root of the polynomial $Q(t) = t^3 - kt - k^2$.

1

- (iii) Hence, find the value(s) of k such that $P(x)$ has its non-real roots lying on the unit circle.

1

- (iv) For those values of k from (iii), does the real root also lie on the unit circle? Justify your answer.

1

- (c) For any positive integer n , define the integral

$$I_n = \int_0^1 \frac{x^n}{\sqrt{1+x^2}} dx.$$

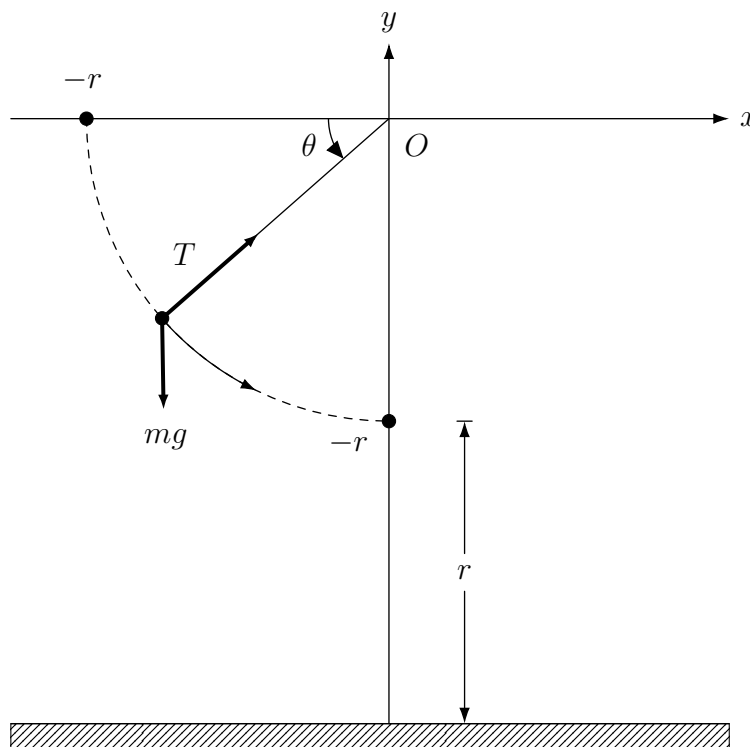
Show that $I_n = \frac{\sqrt{2} - (n-1)I_{n-2}}{n}$ for $n \geq 2$.

3

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- (a) A string of length r is held at fixed point O with an object of mass m attached to the other end. Let O be the origin of the number plane, such that the object is initially at rest at the point $(-r, 0)$ and begins moving anticlockwise along the arc of a circle. Suppose that the fixed point of the string is $2r$ units above the ground.



Let θ be the angular displacement of the object at time t relative to its initial position. At a given time, the object is subjected to a tension force of T and gravitational force mg .

- (i) Let v be the tangential velocity of the object at time t . By resolving the forces into tangential and normal components, show that 4

$$v^2 = 2gr \sin \theta.$$

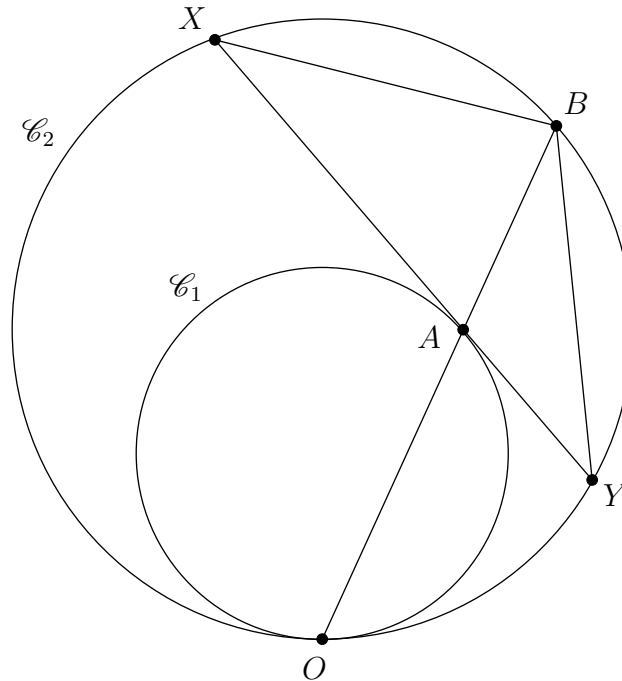
- (ii) When the object reaches the point $(0, -r)$, the string is cut so that the object is released at a horizontal velocity to fall to the ground. Deduce that the object has speed $\sqrt{2gr}$ at release. 1

- (iii) Hence, show that the object hits the ground at the point $(2r, -2r)$. 2

Question 13 continues on page 10

Question 13 (continued)

- (b) The diagram below shows a smaller circle $\mathcal{C}_1$ inside and tangential to a larger circle $\mathcal{C}_2$ at point O . A line through O meets $\mathcal{C}_1$ and $\mathcal{C}_2$ at A and B respectively. From A , a tangent is drawn to intersect $\mathcal{C}_2$ at X and Y . 4



Show that $\triangle BXY$ is isosceles.

- (c) Let $f(x) = 1 + x^r - (1 + x)^r$ for $0 \leq r \leq 1$ and $x \geq 0$.

(i) Show that $f(x) \geq 0$. 2

(ii) Hence, show that for any real values a and b 2

$$|a + b|^r \leq |a|^r + |b|^r.$$

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- (a) A particle of mass m was originally intended to move in simple harmonic motion at time t according to the displacement equation $x = A \cos(nt)$ for positive constants A and n .

However, once the particle was released from rest it experienced a resistance force of $2mnv$, where v is the velocity of the particle.

- (i) Explain why the acceleration equation of the particle with the resistance is

1

$$\ddot{x} = -n^2x - 2nv.$$

- (ii) Let $y = v + nx$. Show that $\dot{y} + ny = 0$.

1

- (iii) Hence show that the displacement equation of the particle is given by

3

$$x = A(1 + nt)e^{-nt}.$$

Hint: Consider the fact that $\frac{d}{dx}(e^x f(x)) = e^x(f(x) + f'(x))$.

- (iv) Sketch the displacement equation of the particle and describe its behaviour over time.

2

- (b) Consider the hyperbola $\mathcal{H}$ defined by the equation

$$y = \frac{x}{\sqrt{3}} + \frac{\sqrt{3}}{x}.$$

It can be shown that $y = \frac{x}{\sqrt{3}}$ is an asymptote of $\mathcal{H}$. (**Do NOT prove this**)

- (i) Sketch the graph of $\mathcal{H}$.

1

- (ii) Find the points intersection of $\mathcal{H}$ and the line $y = x\sqrt{3}$.

2

- (iii) Suppose that $\mathcal{H}$ is rotated clockwise about the origin by $\frac{\pi}{3}$. Using part (ii), or otherwise, show that the equation of the rotated curve is

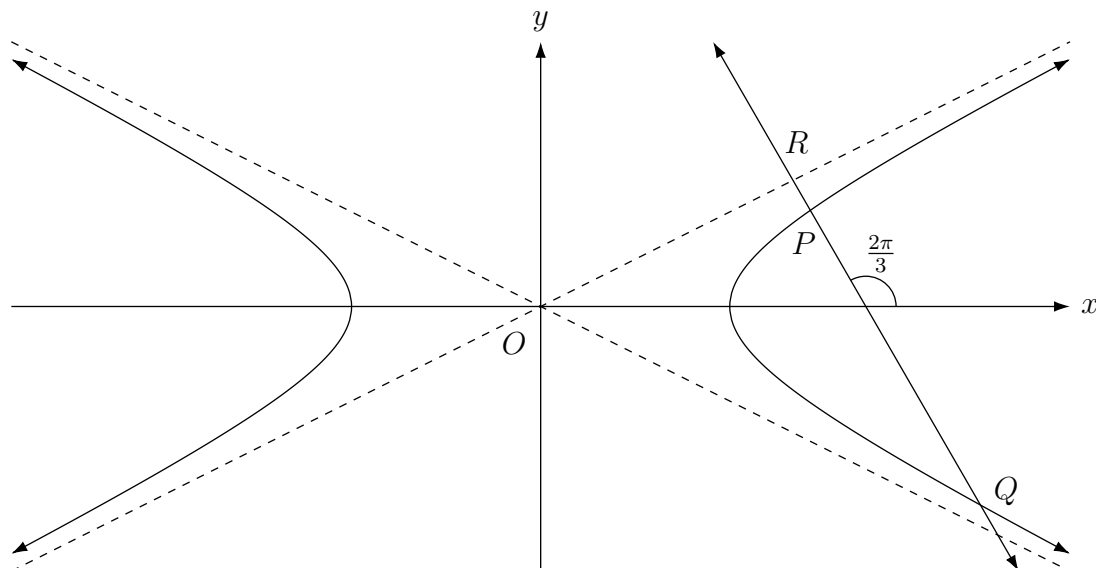
2

$$\frac{x^2}{6} - \frac{y^2}{2} = 1.$$

Question 14 continues on page 12

Question 14 (continued)

- (c) The diagram below shows a hyperbola with the equation $\frac{x^2}{6} - \frac{y^2}{2} = 1$.



The chord PQ on the positive branch of the hyperbola intersects the positive x -axis at an angle of $\frac{2\pi}{3}$. Chord PQ is produced to intersect the positive asymptote $y = \frac{x}{\sqrt{3}}$ at R . Let h be the length of OR .

The region enclosed by the chord and the hyperbola is rotated about the positive asymptote $y = \frac{x}{\sqrt{3}}$ to form a solid of revolution with volume V .

Using the result in part (b)(iii), or otherwise, show that

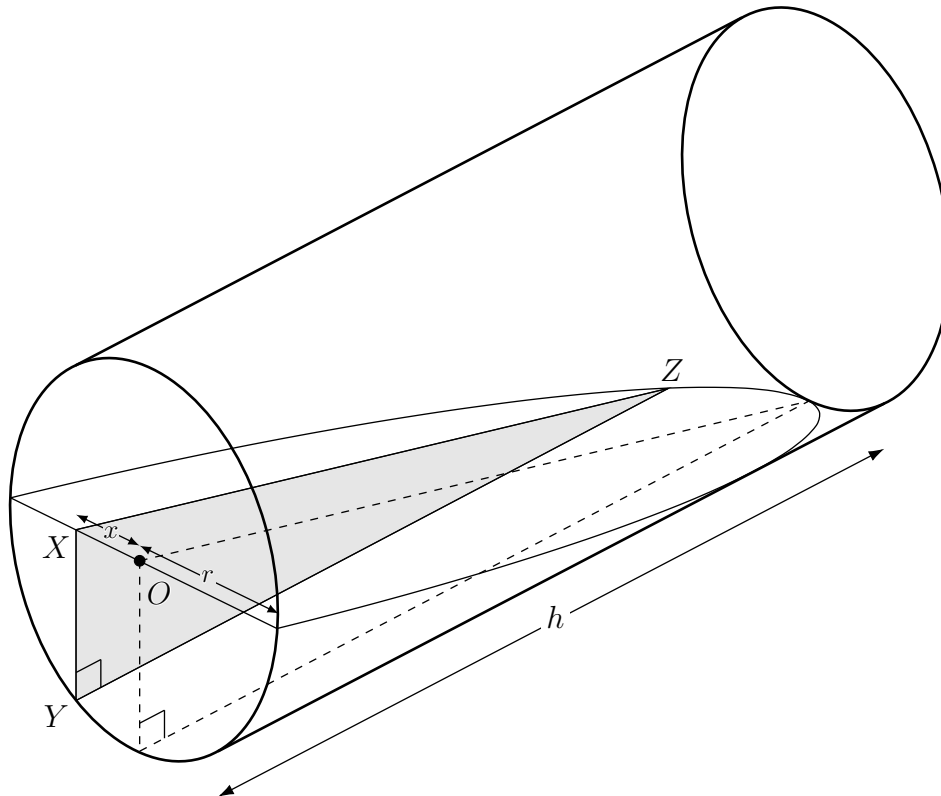
3

$$V = \pi(h^2 - 4)^{\frac{3}{2}}.$$

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

- (a) The diagram below shows a cylinder with height h and radius r , which is filled with water. The cylinder is tilted, spilling any excess water out of it so that the water's surface is level with the centre of cylinder's base.



Cross-sectional slices taken perpendicular to the base are right-angled triangles, and the water surface forms a semi-ellipse. Let $OX = x$.

- | | | |
|-------|---|----------|
| (i) | Find the equation of the semi-ellipse in terms of h and r . | 1 |
| (ii) | Show that the area of $\triangle XYZ$ is $\frac{h}{2r}(r^2 - x^2)$. | 3 |
| (iii) | Hence, find the proportion of the cylinder that is filled with water. | 2 |

Question 15 continues on page 14

Question 15 (continued)

- (b) Let $f(x)$ and $g(x)$ be polynomials such that

$$(1 + ix)^n = f(x) + ig(x),$$

where n is a positive integer.

- (i) Show that

1

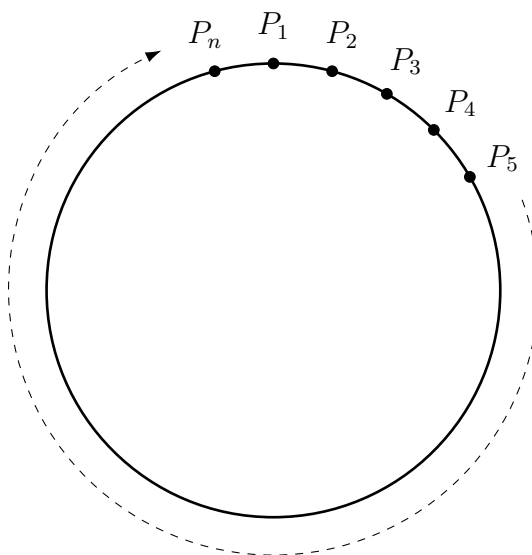
$$(a - bi)(1 + ix)^n + (a + bi)(1 - ix)^n = 2(af(x) + bg(x)),$$

where a and b are any non-zero real numbers.

- (ii) Hence, or otherwise, show that the polynomial $af(x) + bg(x)$ only has real roots.

3

- (c) There are n people sitting in a circle labelled P_1 to P_n as shown in the diagram below.



Every second person is removed from the table in a clockwise direction, initially removing P_2 , until eventually there is only one person remaining on the table.

- (i) Suppose there are 2^m people in the circle labelled P_1 to P_{2^m} , where m is a positive integer. Use mathematical induction to prove that P_1 always wins.

3

- (ii) Hence, find the labelled position of the remaining person if the circle contains 2018 people.

2

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- (a) Elle draws a card from a bag containing cards labelled $1, 2, 3, \dots, n$. The rules are as follows.

- If she draws the lowest number ‘1’, she wins the game.
- If she draws the highest number ‘ n ’ in the bag, she loses the game.
- If she draws any other number, she returns the card that she drew and then adds another card labelled $n + 1$ to the bag and redraws.

This process repeats until eventually Elle wins or loses. Suppose the bag begins with only three cards labelled 1, 2 and 3.

- (i) Show that the probability of Elle winning the game on her k^{th} turn is **2**

$$\frac{2}{k(k+1)(k+2)}.$$

- (ii) Hence, or otherwise, show that **1**

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k(k+1)(k+2)} = \frac{1}{4}.$$

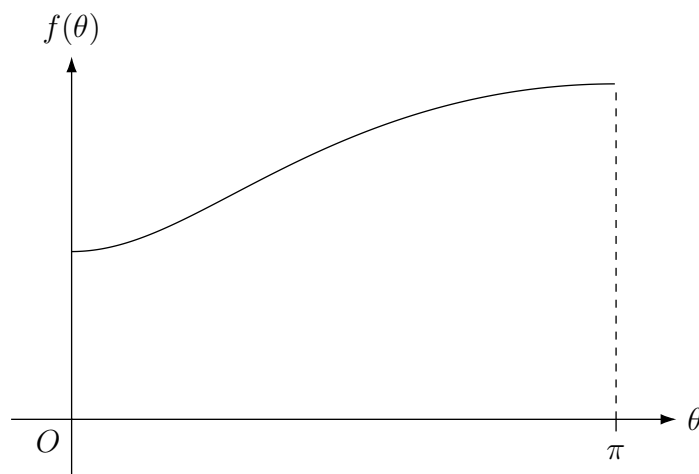
- (b) (i) Find the roots of $P(z) = z^{2n} - 1$ in modulus-argument form. **1**

- (ii) Hence, express $\frac{z^{2n} - 1}{z^2 - 1}$ as the product of real quadratic factors. **1**

Question 16 continues on page 16

Question 16 (continued)

- (c) The diagram below shows the graph of $f(\theta) = \ln(a^2 - 2a \cos \theta + 1)$, where $0 \leq \theta \leq \pi$ and $a > 1$.



- (i) Use part (b) to show that

2

$$\frac{\pi}{n} \sum_{k=1}^n f\left(\frac{k\pi}{n}\right) = R + \frac{\pi}{n} \ln\left(\frac{a+1}{a-1}\right),$$

where $R = \frac{\pi}{n} \ln(a^{2n} - 1)$.

- (ii) Show that

1

$$\frac{\pi}{n} \sum_{k=0}^{n-1} f\left(\frac{k\pi}{n}\right) = R + \frac{\pi}{n} \ln\left(\frac{a-1}{a+1}\right).$$

- (iii) Define the integral

$$I(a) = \int_0^\pi \ln(a^2 - 2a \cos \theta + 1) d\theta.$$

Explain why

1

$$R + \frac{\pi}{n} \ln\left(\frac{a-1}{a+1}\right) < I(a) < R + \frac{\pi}{n} \ln\left(\frac{a+1}{a-1}\right).$$

Question 16 continues on page 17

Question 16 (continued)

- (iv) It can be shown that for $x > 1$ **2**

$$\ln \left(\frac{x}{x-1} \right) < \frac{1}{x-1} \quad (\text{Do NOT prove this})$$

Use this result to show that

$$2\pi \ln a - \frac{\pi}{n} \left(\frac{1}{a^{2n} - 1} \right) < R < 2\pi \ln a.$$

- (v) Show that $I(a) = 2\pi \ln a$. **1**

- (d) Define similarly the same integral from part (c)

$$I(b) = \int_0^\pi \ln(b^2 - 2b \cos \theta + 1) d\theta,$$

but for all $b > 0$, $b \neq 1$.

- (i) Show that $I(b) = 2\pi \ln b + I\left(\frac{1}{b}\right)$. **1**

- (ii) Use part (c) to deduce that **2**

$$I(b) = \begin{cases} 2\pi \ln b, & \text{for } b > 1 \\ 0, & \text{for } 0 < b < 1 \end{cases}$$

End of paper